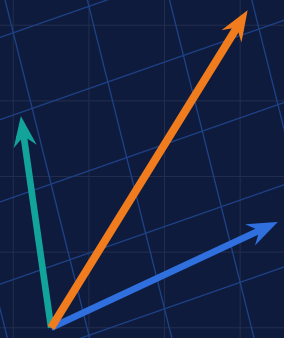
ZMathLab • Linear Algebra

Lecture 3

Matrices and Matrix Operations

From a table of numbers in your daily life to the five ways of multiplying matrices, the transpose and the trace.



$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Muhammad Zubair

BS Linear Algebra • Notes with solved examples

Based on Anton & Rorres §1.3 and Williams §2.1

How to use these notes

Every idea in this lecture comes in the same order: a **story** (why we need it), a **definition** (exactly what it is), **in simple words** (the same thing without symbols), **pictures**, and **examples** where every single step is written. At the end of each part there is a box called *Where is this used?* and, at the end of the notes, a full section on the **mistakes students make**.



Colour code used in every picture. **Blue** = the original object, **teal** = the second input, **orange** = the result, **gold** = the part we are looking at right now (a row, a column, the diagonal).

Book references. Examples marked Anton Ex. 5 come from *Elementary Linear Algebra* (Anton & Rorres), Section 1.3; examples marked Williams come from *Linear Algebra with Applications* (Gareth Williams), Section 2.1. The numbers are the same as in the books, so you can open the book and follow along. Every number in these notes was checked by the Python file `lecture3_matrices.py`.

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1 The story: why do we need matrices?

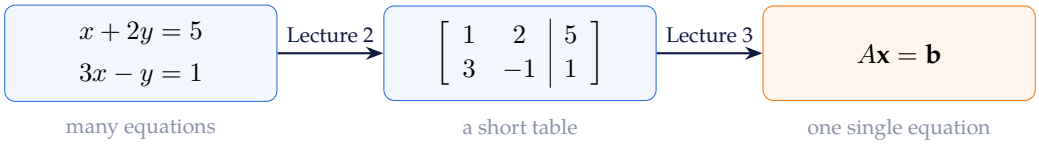
Story: one teacher, forty students, five subjects

Imagine you are a class teacher. You have 40 students and 5 subjects. That is 200 marks. You could write them as 200 separate numbers on a page, but nobody does that. You draw a *table*: one row for each student, one column for each subject. Suddenly you can answer questions in seconds: *Who topped Maths?* (look down a column). *What is Ali's total?* (add along a row).

Now the principal says: "Add 5 grace marks to everyone in Physics," and later "Convert all marks to percentages." You do not do 200 separate jobs. You do **one** operation on the **whole table**.

That is the whole idea of a matrix: **treat a table of numbers as one object**, and learn how to add, scale and multiply such objects. Mathematicians did exactly this in the 1850s. James Sylvester gave the name *matrix* in 1850 and Arthur Cayley wrote the rules for adding and multiplying them in 1858.

In Lecture 2 we already met a matrix without calling it one: the *augmented matrix* of a linear system. We used it only as a short way to write equations. In this lecture a matrix becomes an object of its own, with its own arithmetic.



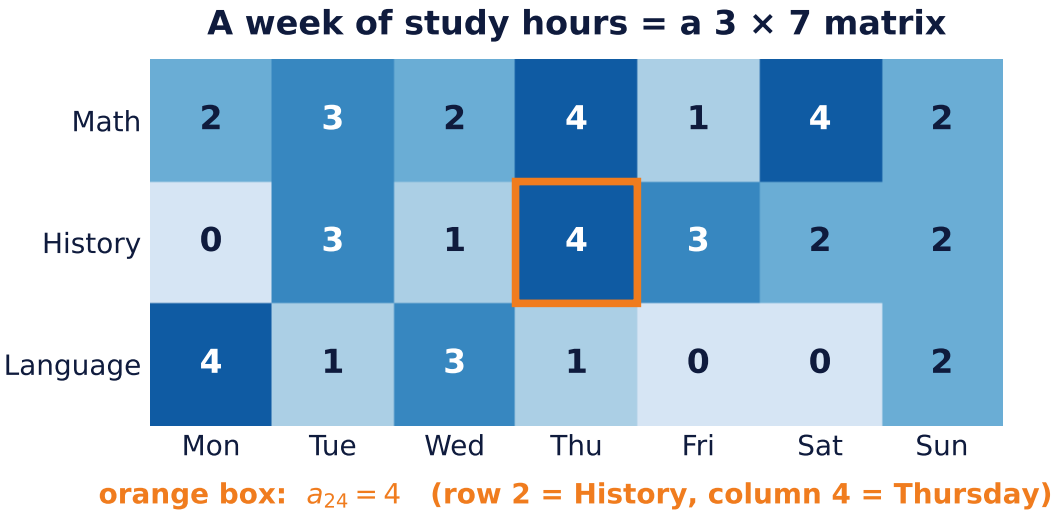
By the end of this lecture you will be able to write a whole system of equations as the single equation $A\mathbf{x} = \mathbf{b}$, which looks exactly like the school equation $ax = b$. That is why matrices are the **core language of Linear Algebra**: they let us treat many numbers, and many equations, as one.

2 Matrices in your daily life

Before any definition, look around you. Matrices are everywhere; we just call them tables.

1. Your study timetable

A student wrote down how many hours they studied each subject during one week Anton §1.3 :



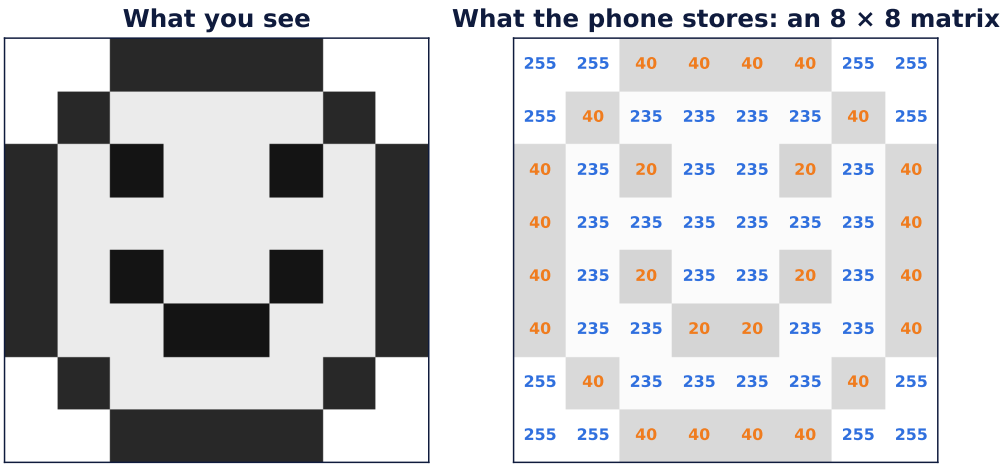
Remove the headings and keep the numbers. What is left is a matrix with **3 rows** (subjects) and **7 columns** (days):

$$\begin{bmatrix} 2 & 3 & 2 & 4 & 1 & 4 & 2 \\ 0 & 3 & 1 & 4 & 3 & 2 & 2 \\ 4 & 1 & 3 & 1 & 0 & 0 & 2 \end{bmatrix}$$

Adding along row 1 gives the total Maths hours, $2+3+2+4+1+4+2 = 18$. Adding down column 4 gives the total hours on Thursday, $4 + 4 + 1 = 9$.

2. The photo on your phone

A black-and-white photo is stored as a grid of numbers: each number is the brightness of one tiny square (pixel), from 0 (black) to 255 (white). The phone does not store a "face"; it stores a matrix.



When you press *brightness* +, the phone adds a number to every entry. When you increase *contrast*, it multiplies every entry by a number. Those are exactly the matrix operations "add" and "scalar multiple" that we learn below. A colour photo is three such matrices (red, green, blue).

3. A cricket scorecard and a shop register

	Runs	Balls	Sixes
Babar	78	61	2
Rizwan	45	40	1
Fakhar	12	9	1

$$\Downarrow \quad \begin{bmatrix} 78 & 61 & 2 \\ 45 & 40 & 1 \\ 12 & 9 & 1 \end{bmatrix}$$

	Rice	Sugar	Oil
Shop 1	12	5	8
Shop 2	20	9	4

$$\Downarrow \quad \begin{bmatrix} 12 & 5 & 8 \\ 20 & 9 & 4 \end{bmatrix}$$

In simple words

A matrix is a table where we have agreed what the rows mean and what the columns mean, and then thrown away the labels so we can do arithmetic with the numbers.

3 Where are matrices used?

Games & graphics
Every turn, zoom or jump of a character on screen is a matrix multiplying the coordinates of its points, thousands of times per second.

Artificial intelligence
A neural network (like the one behind ChatGPT or face unlock) is mainly huge matrices of "weights" multiplied by input vectors.

Google search
Google ranks web pages using one gigantic matrix of "which page links to which page".

Excel & data science
Every spreadsheet is a matrix. Data science is mostly matrix operations on such tables.

Engineering & economics
Electric circuits, bridges, traffic flow and the economy of a whole country are written as $A\mathbf{x} = \mathbf{b}$.

Secret codes
Messages can be locked by multiplying by a matrix and unlocked with its inverse (Lecture 5).

Why matrices are the heart of Linear Algebra

1. **Short to write.** 100 equations in 100 unknowns become one line: $A\mathbf{x} = \mathbf{b}$.
2. **Easy to compute.** Computers are built to do matrix arithmetic extremely fast (that is what a graphics card does).
3. **One idea, many uses.** The same rules work for marks, pixels, prices, forces and probabilities. Learn the rules once, use them everywhere.

4 What is a matrix?

Definition (Anton Def. 1)

A **matrix** is a rectangular array of numbers. The numbers in the array are called the **entries** (or elements) of the matrix.

In simple words

A matrix is a rectangle filled with numbers. Every number sits in a fixed row and a fixed column, like seats in a classroom.

4.1 Size (order) of a matrix

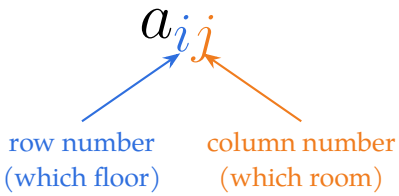
The **size** of a matrix is written **rows** $\times$ **columns**, read "m by n". Rows are horizontal lines, columns are vertical lines. *Rows always come first.*

Example Anton Ex. 1 **: sizes**

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 4 \end{bmatrix}}_{3\times 2}, \quad \underbrace{\begin{bmatrix} 2 & 1 & 0 & -3 \end{bmatrix}}_{1\times 4}, \quad \underbrace{\begin{bmatrix} e & \pi & -\sqrt{2} \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}}_{3\times 3}, \quad \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{2\times 1}, \quad \underbrace{\begin{bmatrix} 4 \end{bmatrix}}_{1\times 1}$$

The first matrix has 3 rows and 2 columns, so its size is 3×2 and it has $3 \cdot 2 = 6$ entries.

4.2 Naming an entry: a_{ij}



The entry in row i and column j of a matrix A is written a_{ij} or $(A)_{ij}$. Think of a building: i is the floor, j is the room on that floor. A general 3×4 matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}.$$

Example Anton §1.3 **: reading entries**

For $A = \begin{bmatrix} 2 & -3 \\ 7 & 0 \end{bmatrix}$: $(A)_{11} = 2, \quad (A)_{12} = -3, \quad (A)_{21} = 7, \quad (A)_{22} = 0.$

In the study-hours matrix, $a_{24} = 4$: row 2 is History and column 4 is Thursday, so the student studied History for 4 hours on Thursday.

Common mistake: mixing up a_{23} and a_{32}

a_{23} means row 2, column 3. a_{32} means row 3, column 2. They are usually different numbers. For $A = \begin{bmatrix} 1 & -2 & -1 \\ 3 & -3 & 4 \\ 2 & 7 & 5 \end{bmatrix}$ Williams we have $a_{23} = 4$ but $a_{32} = 7$.

Capital letters (A, B, C) are used for matrices and small letters (a, b, c, k) for ordinary numbers. Ordinary numbers are called **scalars**.

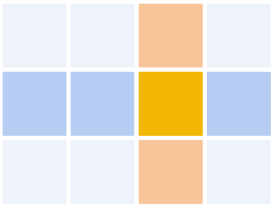
5 Rows, columns and vectors

Definition

A matrix with only **one row** is called a **row vector** (row matrix). A matrix with only **one column** is called a **column vector** (column matrix). Vectors are written with small bold letters:

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \ (1 \times n), \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \ (m \times 1).$$

column 3 = a column vector (3×1): *one subject, all students*



row 2 = a row vector (1×4): *one student, all subjects*

gold cell a_{23} = where they cross

Where is this used?

- A **row vector** is one record: one student's marks, one day's sales, one customer's bill.
- A **column vector** is one feature for everybody, or a point in space: the position of a drone is $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$; the prices of three items form the column $\begin{bmatrix} 200 \\ 150 \\ 400 \end{bmatrix}$.
- A matrix is many rows stacked up, or equally, many columns placed side by side. Keep both pictures in mind; we will use them for multiplication.

6 Writing a big matrix in short form

A general $m \times n$ matrix needs $m \cdot n$ entries:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Instead of writing all of them we write

$$\boxed{A = [a_{ij}]_{m\times n}} \qquad \text{or simply} \qquad A = [a_{ij}].$$

Read it as: "A is the matrix whose entry in row i, column j is a_{ij}; it has m rows and n columns." Use the first form when the size matters.

Example: building a matrix from a rule Anton Exercise 32(a)

Find the 4×4 matrix $A = [a_{ij}]$ with $a_{ij} = i + j$.

Step 1: fill row 1 ($i = 1$)

$a_{11} = 1 + 1 = 2,\ a_{12} = 1 + 2 = 3,\ a_{13} = 1 + 3 = 4,\ a_{14} = 1 + 4 = 5.$

Step 2: do the same for rows 2, 3, 4

$$A = \begin{bmatrix} 1+1 & 1+2 & 1+3 & 1+4 \\ 2+1 & 2+2 & 2+3 & 2+4 \\ 3+1 & 3+2 & 3+3 & 3+4 \\ 4+1 & 4+2 & 4+3 & 4+4 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 \end{bmatrix}.$$

Notice the pattern: numbers are constant along each "anti-diagonal" line, because $i + j$ stays the same when i goes up by 1 and j goes down by 1.

Where is this used?

In programming (Python, MATLAB, Excel) a matrix is stored exactly like this: a name plus a rule or a list, and the program uses `A[i][j]` to reach an entry. The notation $[a_{ij}]$ is how the computer "thinks" about the matrix.

7 Square matrices and diagonals

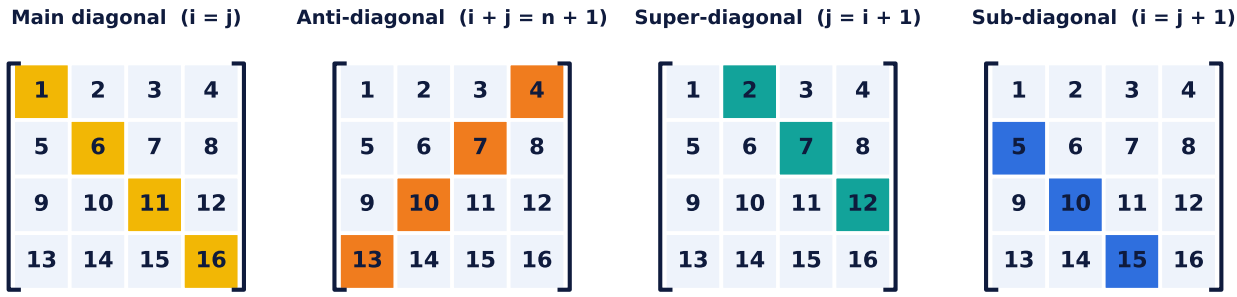
Definition (Anton)

A matrix with n rows and n columns is called a **square matrix of order** n. The entries $a_{11}, a_{22}, \ldots, a_{nn}$ form the **main diagonal** of A.

In simple words

Square means "same number of rows and columns". The main diagonal is the line of entries that goes from the **top-left corner to the bottom-right corner**; on it, the row number equals the column number ($i = j$).

7.1 The main diagonal and the other diagonals



Name	Rule	Entries in a 4×4 matrix
Main diagonal	$i = j$	$a_{11}, a_{22}, a_{33}, a_{44}$ (top-left to bottom-right)
Anti-diagonal (secondary)	$i + j = n + 1$	$a_{14}, a_{23}, a_{32}, a_{41}$ (top-right to bottom-left)
Super-diagonal	$j = i + 1$	a_{12}, a_{23}, a_{34} (just above the main diagonal)
Sub-diagonal	$i = j + 1$	a_{21}, a_{32}, a_{43} (just below the main diagonal)
Above / below the diagonal	$i < j$ / $i > j$	the upper triangle / the lower triangle

Example Williams §2.1

$A = \begin{bmatrix} 1 & -2 & -1 \\ 3 & -3 & 4 \\ 2 & 7 & 5 \end{bmatrix}$ is square of order 3. Its main diagonal is $a_{11} = 1,\ a_{22} = -3,\ a_{33} = 5$. Its anti-diagonal is $a_{13} = -1,\ a_{22} = -3,\ a_{31} = 2$.

Common mistake: "a rectangle has a main diagonal"

The main diagonal is defined only for square matrices. A 2×3 matrix such as $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ has **no main diagonal** in our definition Anton TF (a). (Some books talk about a_{11}, a_{22} for a rectangle, but trace and diagonal matrices need a square.)

7.2 Special matrices built from the diagonal Williams

$$\underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{zero matrix } O_{2\times 3}} \qquad \underbrace{\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 7 \end{bmatrix}}_{\text{diagonal matrix}} \qquad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{identity } I_3} \qquad \underbrace{\begin{bmatrix} 2 & 5 & 1 \\ 0 & 3 & 8 \\ 0 & 0 & 6 \end{bmatrix}}_{\text{upper triangular}}$$

- **Zero matrix** O: every entry is 0. It behaves like the number 0: $A + O = A$.
- **Diagonal matrix**: square, and every entry off the main diagonal is 0.
- **Identity matrix** I_n: diagonal with all 1's. It behaves like the number 1: $AI = IA = A$ ("do nothing").
- **Upper / lower triangular**: all zeros below / above the main diagonal. These are exactly what Gaussian elimination produces in Lecture 2.

8 Equal matrices

Definition (Anton Def. 2)

Two matrices are **equal** if they have the **same size** and their **corresponding entries are equal**. In symbols, $A = B$ means $a_{ij} = b_{ij}$ for every i and j.

In simple words

Two matrices are equal only if they are "photocopies": same shape, and the same number in every seat.

Example Anton Ex. 2

$$A = \begin{bmatrix} 2 & 1 \\ 3 & x \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \end{bmatrix}.$$

A and B

Both are 2×2. Compare seat by seat: $2 = 2$, $1 = 1$, $3 = 3$, and $x = 5$. So $A = B$ exactly when $x = 5$.

A and C

A is 2×2 and C is 2×3. Different sizes, so $A \neq C$ for *every* value of x. (The extra zero column does not "disappear".)

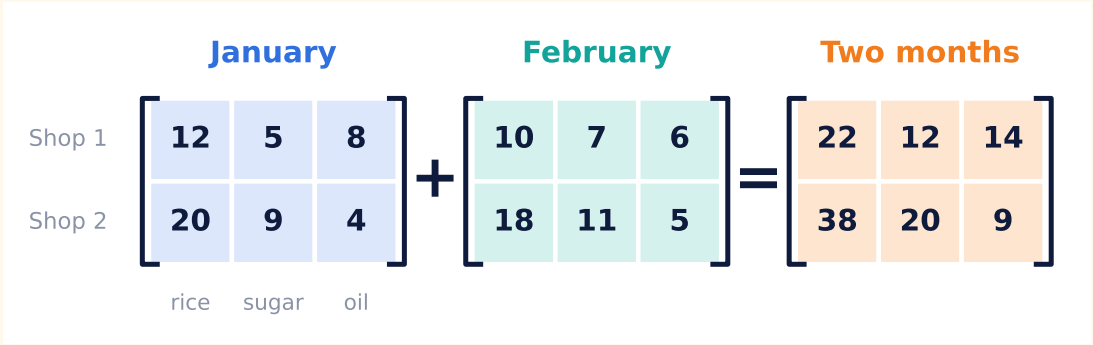
Where is this used?

One matrix equation is many ordinary equations at once. Setting $\begin{bmatrix} a-b & b+a \\ 3d+c & 2d-c \end{bmatrix} = \begin{bmatrix} 8 & 1 \\ 7 & 6 \end{bmatrix}$ gives four equations $a - b = 8,\ a + b = 1,\ 3d + c = 7,\ 2d - c = 6$ Anton Exercise 24, solved in the exercise file.

9 Sum and difference

Story: two months of sales

A shopkeeper has two shops and sells rice, sugar and oil (bags). January and February sales are two 2×3 matrices. "How much did each shop sell of each item in the two months?" is answered by adding the matrices, *seat by seat*:



The 22 in the corner means: Shop 1 sold $12 + 10 = 22$ bags of rice in two months.

Definition (Anton Def. 3)

If A and B have the **same size**, the **sum** $A+B$ is obtained by adding corresponding entries and the **difference** $A-B$ by subtracting corresponding entries:

$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (A-B)_{ij} = a_{ij} - b_{ij}.$$

Matrices of different sizes **cannot** be added or subtracted.

Example Anton Ex. 3

$$A = \begin{bmatrix} 2 & 1 & 0 & 3 \\ -1 & 0 & 2 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -4 & 3 & 5 & 1 \\ 2 & 2 & 0 & -1 \\ 3 & 2 & -4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}.$$

$A+B$ **(both 3×4, so it is defined)**

$$A+B = \begin{bmatrix} 2+(-4) & 1+3 & 0+5 & 3+1 \\ -1+2 & 0+2 & 2+0 & 4+(-1) \\ 4+3 & -2+2 & 7+(-4) & 0+5 \end{bmatrix} = \begin{bmatrix} -2 & 4 & 5 & 4 \\ 1 & 2 & 2 & 3 \\ 7 & 0 & 3 & 5 \end{bmatrix}.$$

$A-B$

$$A-B = \begin{bmatrix} 2-(-4) & 1-3 & 0-5 & 3-1 \\ -1-2 & 0-2 & 2-0 & 4-(-1) \\ 4-3 & -2-2 & 7-(-4) & 0-5 \end{bmatrix} = \begin{bmatrix} 6 & -2 & -5 & 2 \\ -3 & -2 & 2 & 5 \\ 1 & -4 & 11 & -5 \end{bmatrix}.$$

$A+C$, $B+C$, $A-C$, $B-C$

C is 2×2 but A, B are 3×4. Sizes differ, so all four are **undefined**.

Common mistake: subtracting a negative

In $A-B$ the entry $4-(-1)=5$, not 3. Write the brackets every time, exactly as above.

10 Scalar multiple

Story: prices go up by 10%

Last month the prices (in rupees) of rice, sugar and oil were the column $\mathbf{p} = \begin{bmatrix} 200 \\ 150 \\ 400 \end{bmatrix}$. This month everything is 10% dearer. The new prices are $1.1\,\mathbf{p} = \begin{bmatrix} 1.1(200) \\ 1.1(150) \\ 1.1(400) \end{bmatrix} = \begin{bmatrix} 220 \\ 165 \\ 440 \end{bmatrix}$. One number multiplied the whole list. On your phone, "increase contrast" does the same thing to every pixel.

Definition (Anton Def. 4)

If A is any matrix and c is a scalar, the **scalar multiple** cA is the matrix obtained by multiplying **every** entry of A by c: $(cA)_{ij} = c\,a_{ij}$. The size does not change.

Example Anton Ex. 4

$$A=\begin{bmatrix}2&3&4\\1&3&1\end{bmatrix},\quad B=\begin{bmatrix}0&2&7\\-1&3&-5\end{bmatrix},\quad C=\begin{bmatrix}9&-6&3\\3&0&12\end{bmatrix}.$$

$$2A=\begin{bmatrix}2\cdot2&2\cdot3&2\cdot4\\2\cdot1&2\cdot3&2\cdot1\end{bmatrix}=\begin{bmatrix}4&6&8\\2&6&2\end{bmatrix},\quad (-1)B=\begin{bmatrix}0&-2&-7\\1&-3&5\end{bmatrix},\quad \tfrac{1}{3}C=\begin{bmatrix}3&-2&1\\1&0&4\end{bmatrix}.$$

We write $(-1)B$ as $-B$ (the **negative** of B), and then subtraction is just adding a negative: $A-B=A+(-1)B$ Williams .

In simple words

Adding matrices: seat-by-seat. Scalar multiple: one number touches every seat. In both cases the shape never changes.

11 Multiplying two matrices

11.1 The story: a shopping bill

Story: four months of shopping Anton Exercise 33

Type I items cost \$1, Type II cost \$2, Type III cost \$3. The table shows how many of each type were bought in four months:

$$\underbrace{\begin{bmatrix}3&4&3\\5&6&0\\2&9&4\\1&1&7\end{bmatrix}}_{\text{quantities (months}\times\text{types)}}\underbrace{\begin{bmatrix}1\\2\\3\end{bmatrix}}_{\text{prices}}=\begin{bmatrix}3(1)+4(2)+3(3)\\5(1)+6(2)+0(3)\\2(1)+9(2)+4(3)\\1(1)+1(2)+7(3)\end{bmatrix}=\begin{bmatrix}20\\17\\32\\24\end{bmatrix}\ \text{dollars spent.}$$

To get January's bill you go **along the January row** and **down the price column**, multiply pair by pair, and add. That "row times column, then add" is exactly matrix multiplication. Nobody invented it to make life hard; it is how every bill in the world is calculated.

11.2 Definition and the row–column rule

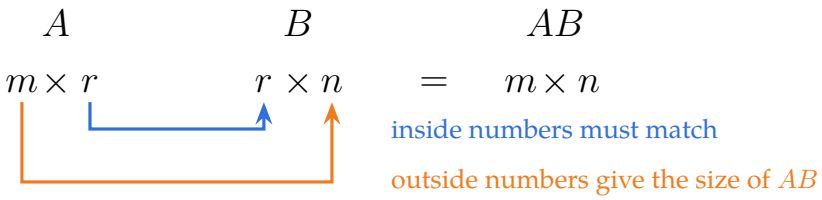
Definition (Anton Def. 5)

If A is $m\times r$ and B is $r\times n$, the **product** AB is the $m\times n$ matrix whose entry in row i, column j is found by taking row i of A and column j of B, multiplying corresponding entries, and adding:

$$\boxed{(AB)_{ij}=a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{ir}b_{rj}}$$

This formula is called the **row–column rule**.

11.3 When is the product defined, and what is its size?



In simple words

Write the sizes side by side. If the two **inside** numbers are equal, you may multiply; the two **outside** numbers are the size of the answer. In words: *the number of columns of A must equal the number of rows of B*, because every row of A has to "meet" every column of B pair by pair.

Example Anton Ex. 6**: which products are defined?**

Suppose A is 3×4, B is 4×7, C is 7×3.

Product	sizes	inside equal?	result
AB	$(3 \times \mathbf{4})(\mathbf{4} \times 7)$	yes	3×7
BC	$(4 \times \mathbf{7})(\mathbf{7} \times 3)$	yes	4×3
CA	$(7 \times \mathbf{3})(\mathbf{3} \times 4)$	yes	7×4
AC	$(3 \times 4)(7 \times 3)$	$4 \neq 7$	undefined
CB	$(7 \times 3)(4 \times 7)$	$3 \neq 4$	undefined
BA	$(4 \times 7)(3 \times 4)$	$7 \neq 3$	undefined

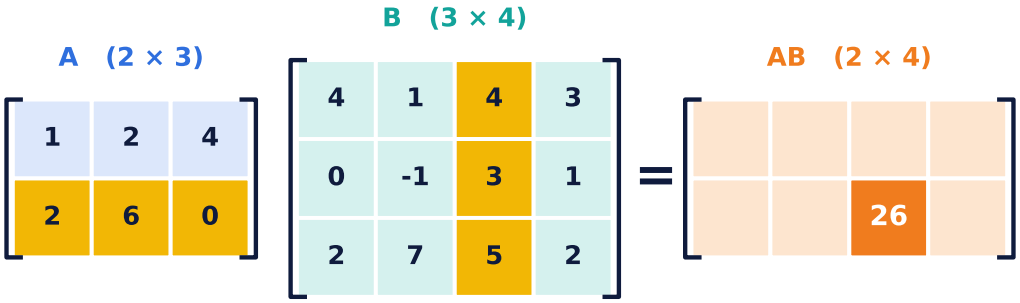
Example Anton Ex. 5**: a full product, every entry**

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \ (2 \times 3), \qquad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} \ (3 \times 4).$$

Step 1: check the size

$(2 \times \mathbf{3})(\mathbf{3} \times 4)$: inside numbers match, so AB exists and is 2×4 (8 entries to find).

Step 2: one entry, slowly. Entry (2,3) = row 2 of A with column 3 of B



row 2 of A · column 3 of B = (2)(4) + (6)(3) + (0)(5) = 26

Step 3: all eight entries

Row 1 of A with each column of B

$(AB)_{11} = (1)(4) + (2)(0) + (4)(2) = 4 + 0 + 8 = 12$
$(AB)_{12} = (1)(1) + (2)(-1) + (4)(7) = 1 - 2 + 28 = 27$
$(AB)_{13} = (1)(4) + (2)(3) + (4)(5) = 4 + 6 + 20 = 30$
$(AB)_{14} = (1)(3) + (2)(1) + (4)(2) = 3 + 2 + 8 = 13$

Row 2 of A with each column of B

$(AB)_{21} = (2)(4) + (6)(0) + (0)(2) = 8 + 0 + 0 = 8$
$(AB)_{22} = (2)(1) + (6)(-1) + (0)(7) = 2 - 6 + 0 = -4$
$(AB)_{23} = (2)(4) + (6)(3) + (0)(5) = 8 + 18 + 0 = 26$
$(AB)_{24} = (2)(3) + (6)(1) + (0)(2) = 6 + 6 + 0 = 12$

Step 4: write the answer

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}.$$

And BA? $(3 \times \mathbf{4})(\mathbf{2} \times 3)$: $4 \neq 2$, so BA does not exist at all.

Example Williams Ex. 3**: one entry without the whole product**

$A = \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -7 & 3 & 2 \\ 5 & 0 & 1 \end{bmatrix}$, $C = AB$. Find c_{23}.

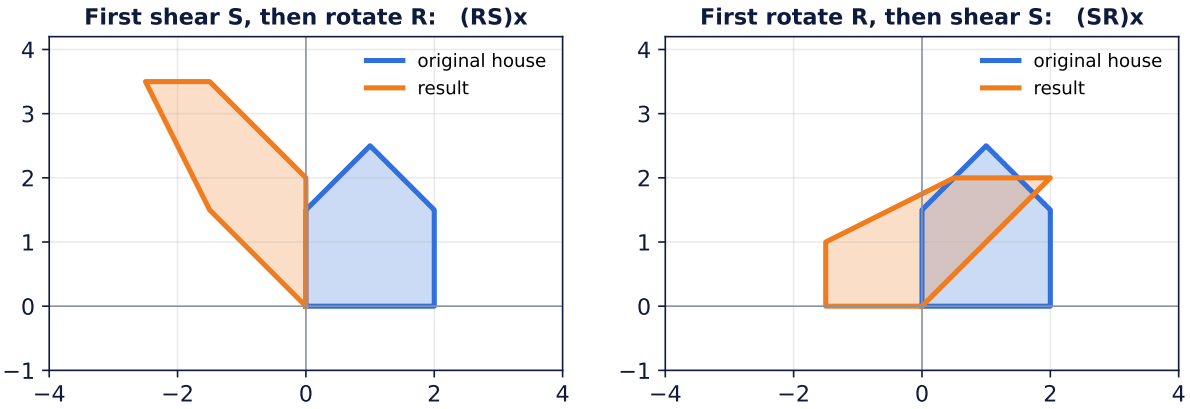
c_{23} uses **row 2 of** A and **column 3 of** B: $c_{23} = \begin{bmatrix} -3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (-3)(2) + (4)(1) = -6 + 4 = -2.$

11.4 What does a product do? A picture

A 2×2 matrix can act on every point (x, y) of a picture: the point $\begin{bmatrix} x \\ y \end{bmatrix}$ goes to $A \begin{bmatrix} x \\ y \end{bmatrix}$. Take $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (turn by 90°) and $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (push sideways, a "shear"). The product RS means *first* S, *then* R; SR means *first* R, *then* S:

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \qquad SR = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}.$$

Same two moves, different order, different house: RS ≠ SR



Putting on socks and then shoes is not the same as shoes and then socks. In the same way AB **and** BA

are usually different. This is the single most important difference between matrices and ordinary numbers.

Example Williams Ex. 2: $CD \neq DC$

$$C = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}, \; D = \begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix}.$$

$$CD = \begin{bmatrix} (1)(4)+(2)(2) & (1)(-1)+(2)(3) \\ (-3)(4)+(0)(2) & (-3)(-1)+(0)(3) \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -12 & 3 \end{bmatrix},$$

$$DC = \begin{bmatrix} (4)(1)+(-1)(-3) & (4)(2)+(-1)(0) \\ (2)(1)+(3)(-3) & (2)(2)+(3)(0) \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ -7 & 4 \end{bmatrix}.$$

Both products exist, both are 2×2, and still $CD \neq DC$.

12 Partitioned (block) matrices

Story: cutting a big cake

A school has 2000 students and 10 subjects: a 2000×10 marks matrix. The boys' section and the girls' section, science and arts subjects... It is natural to **cut the big table into smaller tables** (blocks) and work with the blocks. Computers do this all the time: a huge matrix is cut into blocks that fit in memory, and blocks full of zeros are simply skipped.

Definition

A matrix is **partitioned** when we draw horizontal and/or vertical lines between some of its rows and columns. The smaller matrices formed are called **submatrices** or **blocks**.

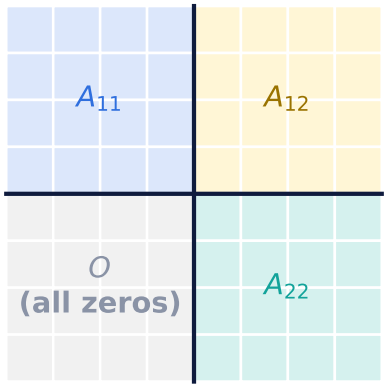
12.1 Three ways to cut the same matrix Anton

$$A = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ \hline a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad A = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \hline a_{21} & a_{22} & a_{23} & a_{24} \\ \hline a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{bmatrix},$$

$$A = \left[\begin{array}{c|c|c|c} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 & \mathbf{c}_4 \end{bmatrix}.$$

The first cuts A into four blocks, the second into its **row vectors**, the third into its **column vectors**. The last two will give us new ways to multiply.

An 8 × 8 matrix seen as a 2 × 2 matrix of 4 × 4 blocks



12.2 Block multiplication

Theorem (block rule)

If the blocks have matching sizes, partitioned matrices can be added and multiplied **as if the blocks were ordinary numbers**. For multiplication, the way we cut the **columns of** A must match the way we cut the **rows of** B:

$$\begin{bmatrix} P & Q \\ R & S \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \begin{bmatrix} PM + QN \\ RM + SN \end{bmatrix}.$$

Example Williams Ex. 5**: block multiplication with every step**

$$A = \left[\begin{array}{c|c} 1 & -1 \\ 3 & 0 \\ \hline 2 & 4 \end{array}\right], \qquad B = \left[\begin{array}{ccc} 1 & 2 & -1 \\ \hline 1 & 3 & 1 \end{array}\right].$$

A is cut into 2 block-columns (sizes 1 and 1), so B must be cut into 2 block-rows (sizes 1 and 1). Name the blocks

$$P = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \; Q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \; R = \begin{bmatrix} 2 \end{bmatrix}, \; S = \begin{bmatrix} 4 \end{bmatrix}; \qquad M = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}, \; N = \begin{bmatrix} 1 & 3 & 1 \end{bmatrix}.$$

Top block: $PM + QN$

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & 2 & -1\end{bmatrix}+\begin{bmatrix}-1\\0\end{bmatrix}\begin{bmatrix}1 & 3 & 1\end{bmatrix}=\begin{bmatrix}1 & 2 & -1\\3 & 6 & -3\end{bmatrix}+\begin{bmatrix}-1 & -3 & -1\\0 & 0 & 0\end{bmatrix}=\begin{bmatrix}0 & -1 & -2\\3 & 6 & -3\end{bmatrix}.$$

Bottom block: $RM + SN$

$$[2]\begin{bmatrix}1 & 2 & -1\end{bmatrix}+[4]\begin{bmatrix}1 & 3 & 1\end{bmatrix}=\begin{bmatrix}2 & 4 & -2\end{bmatrix}+\begin{bmatrix}4 & 12 & 4\end{bmatrix}=\begin{bmatrix}6 & 16 & 2\end{bmatrix}.$$

Put the blocks back together

$$AB=\begin{bmatrix}0 & -1 & -2\\3 & 6 & -3\\6 & 16 & 2\end{bmatrix}.$$

Check with the ordinary row–column rule: $(AB)_{11}=(1)(1)+(-1)(1)=0$ ✓, $(AB)_{32}=(2)(2)+(4)(3)=16$ ✓.

Where is this used?

- **Big data:** a matrix too big for memory is multiplied block by block.
- **Zero blocks save work:** if a block is O, every product with it is O, so the computer skips it.
- **Parallel computing:** different blocks are sent to different processor cores at the same time.

13 Multiplying by columns and by rows

Cut B into its columns, or A into its rows, and the block rule gives two very useful formulas (Anton, formulas (6)–(9)):

$$\boxed{AB=A\begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n\end{bmatrix}=\begin{bmatrix}A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n\end{bmatrix}}$$

$$\boxed{AB=\begin{bmatrix}\mathbf{a}_1\\\mathbf{a}_2\\\vdots\\\mathbf{a}_m\end{bmatrix}B=\begin{bmatrix}\mathbf{a}_1B\\\mathbf{a}_2B\\\vdots\\\mathbf{a}_mB\end{bmatrix}}$$

In simple words

j-th column of AB = A times the j-th column of B. i-th row of AB = the i-th row of A times B. So if you need only one row or one column of a big product, you never compute the rest.

Example Anton Ex. 7**: one column and one row of the product in Example 5**

Second column of AB = A·(second column of B)

$$\begin{bmatrix}1 & 2 & 4\\2 & 6 & 0\end{bmatrix}\begin{bmatrix}1\\-1\\7\end{bmatrix}=\begin{bmatrix}(1)(1)+(2)(-1)+(4)(7)\\(2)(1)+(6)(-1)+(0)(7)\end{bmatrix}=\begin{bmatrix}27\\-4\end{bmatrix}.$$

First row of AB = (first row of A)·B

$$\begin{bmatrix}1 & 2 & 4\end{bmatrix}\begin{bmatrix}4 & 1 & 4 & 3\\0 & -1 & 3 & 1\\2 & 7 & 5 & 2\end{bmatrix}=\begin{bmatrix}12 & 27 & 30 & 13\end{bmatrix}$$

(the four numbers are $4+0+8,\ 1-2+28,\ 4+6+20,\ 3+2+8$). Both agree with the full product in Example 5.

Example Williams **: a product built column by column**

$A=\begin{bmatrix}2&0\\1&5\end{bmatrix}$, $B=\begin{bmatrix}4&1&3\\0&2&-1\end{bmatrix}$.

$$A\begin{bmatrix}4\\0\end{bmatrix}=\begin{bmatrix}8\\4\end{bmatrix},\quad A\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}2\\11\end{bmatrix},\quad A\begin{bmatrix}3\\-1\end{bmatrix}=\begin{bmatrix}6\\-2\end{bmatrix}\quad\Longrightarrow\quad AB=\begin{bmatrix}8&2&6\\4&11&-2\end{bmatrix}.$$

14 Matrix products as linear combinations

Definition (Anton Def. 6)

If $A_1, A_2, \ldots, A_r$ have the same size and $c_1, \ldots, c_r$ are scalars, then $c_1A_1 + c_2A_2 + \cdots + c_rA_r$ is called a **linear combination** of $A_1, \ldots, A_r$ with **coefficients** $c_1, \ldots, c_r$.

In simple words

A linear combination is a *recipe*: "take 2 spoons of this, 1 spoon of that, 3 spoons of the other, and mix (add)". The coefficients are the spoon counts.

Theorem 1.3.1 Anton

If A is $m \times n$ and $\mathbf{x}$ is $n \times 1$, then $A\mathbf{x}$ is a linear combination of the **columns** of A, and the coefficients are the entries of $\mathbf{x}$:

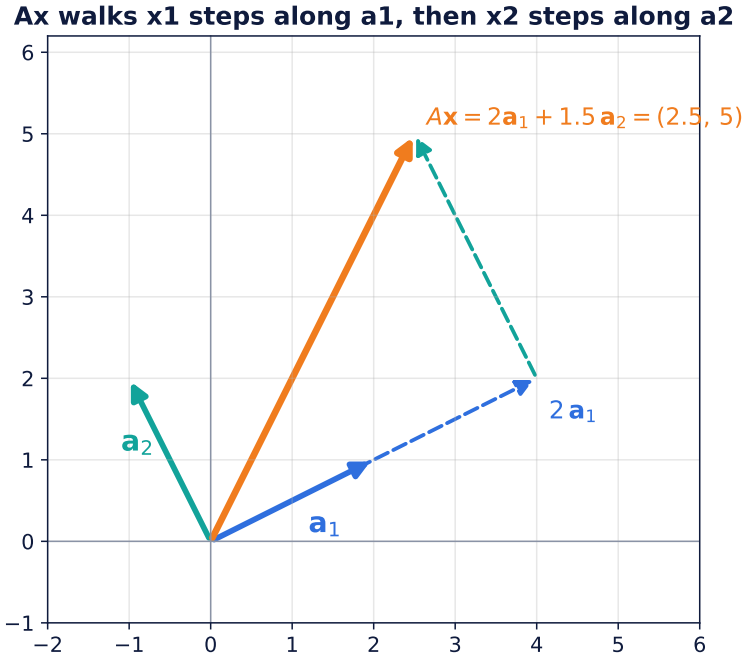
$$A\mathbf{x} = x_1\,\mathbf{c}_1 + x_2\,\mathbf{c}_2 + \cdots + x_n\,\mathbf{c}_n.$$

Example Anton Ex. 8

$$\begin{bmatrix}-1&3&2\\1&2&-3\\2&1&-2\end{bmatrix}\begin{bmatrix}2\\-1\\3\end{bmatrix}=2\begin{bmatrix}-1\\1\\2\end{bmatrix}-1\begin{bmatrix}3\\2\\1\end{bmatrix}+3\begin{bmatrix}2\\-3\\-2\end{bmatrix}=\begin{bmatrix}-2-3+6\\2-2-9\\4-1-6\end{bmatrix}=\begin{bmatrix}1\\-9\\-3\end{bmatrix}.$$

14.1 The picture

In two dimensions the columns of A are two arrows. $A\mathbf{x}$ is where you land if you walk x_1 copies of the first arrow and then x_2 copies of the second:



Here $A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1.5 \end{bmatrix}$: $A\mathbf{x} = 2\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1.5\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 5 \end{bmatrix}$.

Example Anton Ex. 9**: every column of AB is a combination of the columns of A**

With A, B from Example 5, column j of AB uses the entries of column j of B as coefficients:

$$\begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0\begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2\begin{bmatrix} 4 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 27 \\ -4 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 1\begin{bmatrix} 2 \\ 6 \end{bmatrix} + 7\begin{bmatrix} 4 \\ 0 \end{bmatrix},$$

$$\begin{bmatrix} 30 \\ 26 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3\begin{bmatrix} 2 \\ 6 \end{bmatrix} + 5\begin{bmatrix} 4 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 13 \\ 12 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1\begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2\begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

Where is this used?

This view answers the big question of Linear Algebra: *"Can* $\mathbf{b}$ *be built from the columns of* A*?"* If yes, $A\mathbf{x} = \mathbf{b}$ has a solution and $\mathbf{x}$ is the recipe. A colour on your screen is a linear combination of red, green and blue; a smoothie is a linear combination of fruits.

15 Column–row expansion

If A $(m \times r)$ is cut into columns $\mathbf{c}_1, \ldots, \mathbf{c}_r$ and B $(r \times n)$ into rows $\mathbf{r}_1, \ldots, \mathbf{r}_r$, then each product $\mathbf{c}_k\mathbf{r}_k$ is a full $m \times n$ matrix, and Anton (11)

$$\boxed{AB = \mathbf{c}_1\mathbf{r}_1 + \mathbf{c}_2\mathbf{r}_2 + \cdots + \mathbf{c}_r\mathbf{r}_r}$$

In simple words

Instead of "row times column = one number" we do "column times row = a whole layer". Stack (add) the layers and you get AB.

Example Anton Ex. 10

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, \; B = \begin{bmatrix} 2 & 0 & 4 \\ -3 & 5 & 1 \end{bmatrix}.$$

Step 1: columns of A, rows of B

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; \mathbf{c}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}; \quad \mathbf{r}_1 = \begin{bmatrix} 2 & 0 & 4 \end{bmatrix}, \; \mathbf{r}_2 = \begin{bmatrix} -3 & 5 & 1 \end{bmatrix}.$$

Step 2: each layer (column $\times$ row)

$$\mathbf{c}_1\mathbf{r}_1 = \begin{bmatrix} 1\cdot 2 & 1\cdot 0 & 1\cdot 4 \\ 2\cdot 2 & 2\cdot 0 & 2\cdot 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 4 \\ 4 & 0 & 8 \end{bmatrix},$$

$$\mathbf{c}_2\mathbf{r}_2 = \begin{bmatrix} 3(-3) & 3(5) & 3(1) \\ -1(-3) & -1(5) & -1(1) \end{bmatrix} = \begin{bmatrix} -9 & 15 & 3 \\ 3 & -5 & -1 \end{bmatrix}.$$

Step 3: add the layers

$\mathbf{c}_1\mathbf{r}_1$ $\mathbf{c}_2\mathbf{r}_2$ AB

$$\begin{bmatrix} 2 & 0 & 4 \\ 4 & 0 & 8 \end{bmatrix} + \begin{bmatrix} -9 & 15 & 3 \\ 3 & -5 & -1 \end{bmatrix} = \begin{bmatrix} -7 & 15 & 7 \\ 7 & -5 & 7 \end{bmatrix}$$

Step 4: check with the row–column rule

$(AB)_{11} = (1)(2) + (3)(-3) = -7$ ✓, $(AB)_{23} = (2)(4) + (-1)(1) = 7$ ✓.

Where is this used?

Each layer $\mathbf{c}_k\mathbf{r}_k$ is a very simple matrix (every row is a multiple of the same row). Photo compression (SVD) keeps only the few most important layers of a picture and throws the rest away: the photo still looks almost the same but needs far less storage.

Summary: five ways to compute AB Anton

Method	Formula	Best when you need
1. Entry by entry	Definition 5	a product of small matrices by hand
2. Row–column rule	$(AB)_{ij} = \sum_k a_{ik}b_{kj}$	one particular entry
3. Column by column	$AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$	one column; theory of $A\mathbf{x} = \mathbf{b}$
4. Row by row	ith row of $AB = \mathbf{a}_i B$	one row
5. Column–row expansion	$AB = \sum_k \mathbf{c}_k \mathbf{r}_k$	seeing AB as a sum of simple layers

16 A whole linear system as one equation

A system of m equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ \vdots \qquad\qquad & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \quad \Longleftrightarrow \quad \underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}}_{\mathbf{b}}$$

because two matrices are equal exactly when their entries are equal. A is the **coefficient matrix** and $[A \mid \mathbf{b}]$ is the augmented matrix of Lecture 2.

17 The transpose of a matrix

Story: turning the register sideways

A teacher's register has students as rows and subjects as columns. The examination branch wants the same data with *subjects as rows*. Nothing new is calculated; the table is simply turned so that rows become columns. In Excel this is the button *Paste → Transpose*.

Definition (Anton Def. 7)

The **transpose** of an $m \times n$ matrix A, written A^T, is the $n \times m$ matrix obtained by interchanging rows and columns: the first column of A^T is the first row of A, and so on. In entries:

$$\boxed{(A^T)_{ij} = (A)_{ji}}$$

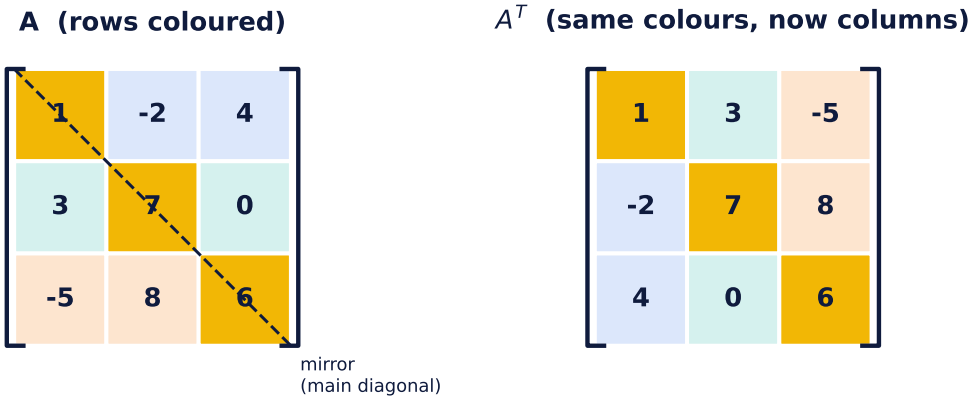
Example Anton Ex. 11

$$B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 5 & 6 \end{bmatrix} \Rightarrow B^T = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 6 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix} \Rightarrow C^T = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \qquad D = [4] \Rightarrow D^T = [4].$$

Sizes: B is 3×2, B^T is 2×3. A row vector becomes a column vector.

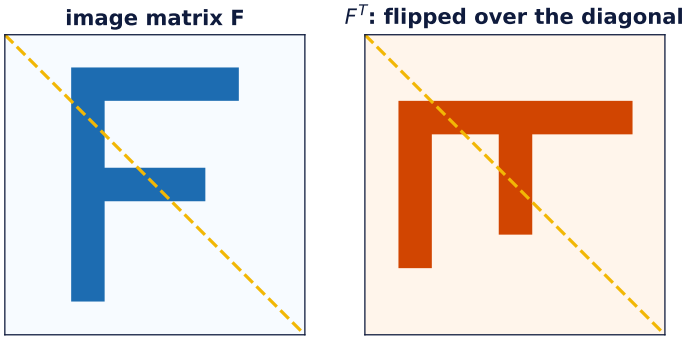
17.1 The picture: a mirror on the main diagonal

For a square matrix, the transpose is a **reflection in the main diagonal**: entries on the diagonal stay where they are, and a_{ij} swaps places with a_{ji} Anton (15) .



$$A = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 7 & 0 \\ -5 & 8 & 6 \end{bmatrix} \longrightarrow A^T = \begin{bmatrix} 1 & 3 & -5 \\ -2 & 7 & 8 \\ 4 & 0 & 6 \end{bmatrix}.$$

Do the same to a picture (a matrix of pixels) and the picture is flipped over its diagonal:



17.2 Rules for the transpose

$(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(kA)^T = kA^T$, $\boxed{(AB)^T = B^T A^T}$ (order reverses!)

Common mistake: $(AB)^T = A^T B^T$

Take $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & 1 \\ 5 & -2 \end{bmatrix}$. Then $XY = \begin{bmatrix} 10 & -3 \\ 20 & -5 \end{bmatrix}$, so $(XY)^T = \begin{bmatrix} 10 & 20 \\ -3 & -5 \end{bmatrix}$. $Y^T X^T = \begin{bmatrix} 0 & 5 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 20 \\ -3 & -5 \end{bmatrix}$ ✓, but $X^T Y^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix}$ ✗. Like socks and shoes: to undo, reverse the order.

18 Homogeneous systems: trivial and nontrivial solutions

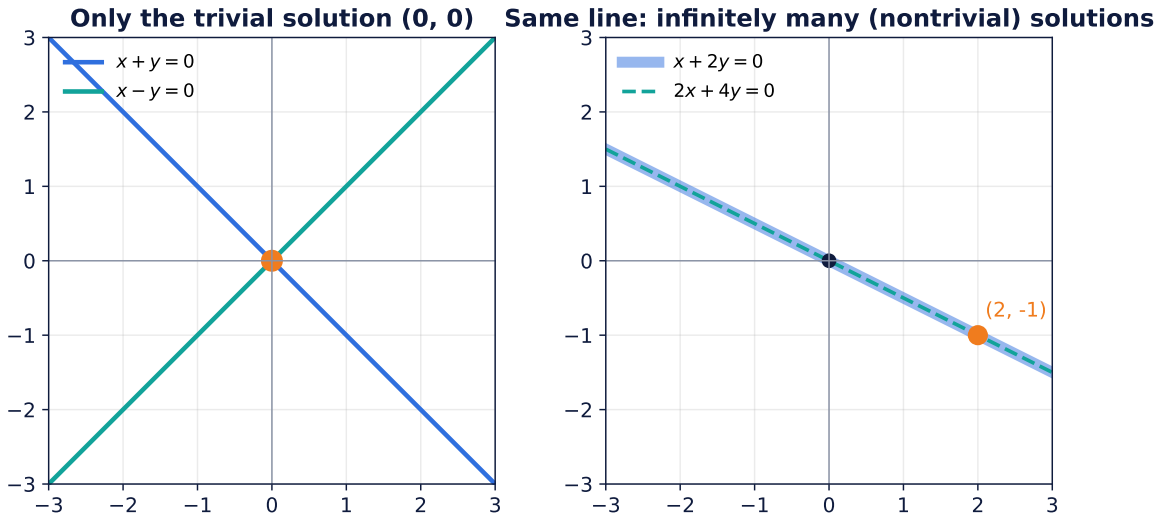
Definition

A linear system is **homogeneous** if every constant on the right side is zero, i.e. it has the form $A\mathbf{x} = \mathbf{0}$.

- $\mathbf{x} = \mathbf{0}$ (all unknowns zero) always works. It is called the **trivial solution**.
- Any other solution (at least one unknown not zero) is a **nontrivial solution**.

In simple words

"Trivial" means boring: of course zero times anything is zero. The interesting question is: *is zero the only answer, or are there others?* There are only two possibilities: **only the trivial solution**, or **infinitely many solutions**. "No solution" can never happen, because **0** always works.



In 2D every equation $ax + by = 0$ is a line **through the origin**. Two different lines meet only at $(0, 0)$: only the trivial solution. If both equations are the same line (one is a multiple of the other), every point on it is a solution.

Example: deciding which case

(a)

$\begin{cases} x + y = 0 \\ x - y = 0 \end{cases}$ Adding: $2x = 0 \Rightarrow x = 0$, then $y = 0$. **Only the trivial solution.**

(b)

$\begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases}$ $R_2 \to R_2 - 2R_1$ gives $0 = 0$. One equation, two unknowns: $y = t$ free, $x = -2t$.

For $t = -1$: $(x, y) = (2, -1)$, a **nontrivial solution**; there are infinitely many.

18.1 Link with linear combinations

By Theorem 1.3.1, $A\mathbf{x} = \mathbf{0}$ says $x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{0}$. So a nontrivial solution is a recipe that mixes the columns of A into nothing. In (b), $2\begin{bmatrix}1\\2\end{bmatrix} - 1\begin{bmatrix}2\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$: the second column is just twice the first (the columns are "dependent").

Theorem (Anton Thm. 1.2.2)

A homogeneous system with **more unknowns than equations** always has infinitely many solutions (so it has nontrivial solutions).

Where is this used?

In chemistry, balancing $x_1\,H_2 + x_2\,O_2 \to x_3\,H_2O$ is a homogeneous system: we look for a *nontrivial* solution ($x_1 = 2, x_2 = 1, x_3 = 2$). The trivial one "no reaction at all" is useless. In engineering, a nontrivial solution of $A\mathbf{x} = \mathbf{0}$ can mean a structure can move without any force: a warning sign.

19 The trace of a matrix

Story: who stayed at home?

Three cities, Lahore, Karachi and Islamabad. A survey records, for one year, how many families (in thousands) moved from city i to city j: that is a 3×3 matrix M. Off the diagonal are the families who *moved*. On the main diagonal, m_{11}, m_{22}, m_{33}, are the families who *stayed in their own city*. Add the diagonal and you get one number for the whole country: how many families did not move. That one number is the **trace** of M. In data science the same idea gives the total spread (total variance) of a whole data set in one number.

Definition (Anton Def. 8)

If A is a **square** matrix, the **trace** of A, written $\mathrm{tr}(A)$, is the sum of the entries on the main diagonal:

$$\mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

The trace is **not defined** if A is not square.

Example Anton Ex. 12

$$B = \begin{bmatrix} -1 & 2 & 7 & 0 \\ 3 & 5 & -8 & 4 \\ 1 & 2 & 7 & -3 \\ 4 & -2 & 1 & 0 \end{bmatrix}$$

$\operatorname{tr}(B) = -1 + 5 + 7 + 0 = 11.$

For a general 3×3: $\operatorname{tr}(A) = a_{11} + a_{22} + a_{33}$.

-1	2	7	0
3	**5**	-8	4
1	2	**7**	-3
4	-2	1	**0**

tr(B) = -1 + 5 + 7 + 0 = 11

19.1 Why is the trace important?

- It squeezes a whole matrix into **one number** that does not change when you transpose: $\operatorname{tr}(A^T) = \operatorname{tr}(A)$ (the diagonal is the mirror line, it does not move).
- It is easy and **linear**: $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ and $\operatorname{tr}(cA) = c \operatorname{tr}(A)$.
- The surprise: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ even though $AB \neq BA$.
- In later lectures you will see that the trace equals the sum of the *eigenvalues* of the matrix; physicists and data scientists use it every day.

Common mistake: $\operatorname{tr}(AB) = \operatorname{tr}(A)\operatorname{tr}(B)$

$P = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: $PQ = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, so $\operatorname{tr}(PQ) = 2 + 3 = 5$. But $\operatorname{tr}(P)\operatorname{tr}(Q) = (1 + 4)(0 + 0) = 0$. Not equal Anton TF (f) .

20 Mistakes students make (and how to avoid them)

Common mistake 1: multiplying entry by entry

$\begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix}$ is **not** $\begin{bmatrix} 4 & -2 \\ -6 & 0 \end{bmatrix}$. The correct product (row times column) is $\begin{bmatrix} 8 & 5 \\ -12 & 3 \end{bmatrix}$ Williams Ex. 2 .

Common mistake 2: assuming $AB = BA$

Even when both exist and have the same size, usually $AB \neq BA$ (Williams Ex. 2: $CD \neq DC$; the house picture). Always keep the order. Never "cancel" or "swap" like with numbers.

Common mistake 3: multiplying when the sizes do not fit

$(2 \times 3)(2 \times 3)$ cannot be multiplied: inside numbers $3 \neq 2$. Check sizes **before** you start, every time.

Common mistake 4: wrong size of the answer

$(3 \times 4)(4 \times 7)$ is 3×7, not 4×4 and not 3×4. Outside numbers give the answer.

Common mistake 5: cancelling in $AC = BC$

With numbers, $ac = bc$, $c \neq 0 \Rightarrow a = b$. With matrices this fails. $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$: $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ but $B \neq C$.

Common mistake 6: "if $AA = O$ then $A = O$"

$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq O$ but $AA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Anton Exercise 30 .

Common mistake 7: $(A + B)^2 = A^2 + 2AB + B^2$

Correct: $(A+B)^2 = (A+B)(A+B) = A^2+AB+BA+B^2$. The school formula needs $AB = BA$, which is usually false.

Common mistake 8: transpose and trace slips

$(AB)^T = B^T A^T$ (order reverses); tr is only for square matrices; $\operatorname{tr}(AB) \neq \operatorname{tr}(A)\operatorname{tr}(B)$ in general.

Common mistake 9: adding a number to a matrix

$A+5$ is **not defined**. You can add a matrix to a matrix of the same size, or multiply a matrix by a number. (Software like Python "broadcasts" $A+5$, but in Linear Algebra it is not an operation.)

21 The whole lecture on one page

Matrix	rectangular array; size $m \times n$ = rows × columns; entry a_{ij} in row i, column j; short form $[a_{ij}]_{m\times n}$
Vectors	row vector $1 \times n$; column vector $m \times 1$
Square, diagonal	$n \times n$; main diagonal $a_{11}, \ldots, a_{nn}$ $(i = j)$; zero, diagonal, identity, triangular matrices
Equality	same size and same entries
$A \pm B$	same size; entry by entry
cA	every entry times c
AB	$(m\times r)(r\times n) = m\times n$; $(AB)_{ij}$ = row i of A · column j of B; usually $AB \neq BA$
Five views	entry, row–column, columns $[A\mathbf{b}_j]$, rows $[\mathbf{a}_i B]$, column–row $\sum \mathbf{c}_k\mathbf{r}_k$
$A\mathbf{x}$	$x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n$ (linear combination of columns)
Transpose	rows ↔ columns; $(A^T)_{ij} = a_{ji}$; $(AB)^T = B^T A^T$
$A\mathbf{x} = \mathbf{0}$	always has $\mathbf{x} = \mathbf{0}$ (trivial); either only that, or infinitely many
Trace	square only; sum of main diagonal; $\mathrm{tr}(AB) = \mathrm{tr}(BA)$

Quick check (answers at the bottom)

1. A is 2×5 and B is 5×3. Is BA defined? What is the size of AB?
2. Write the 3×3 matrix with $a_{ij} = i - j$ and find its trace.
3. True or false: if A and B are 2×2 then $AB = BA$.
4. Does $\begin{cases} x + y + z = 0 \\ x - y + 2z = 0 \end{cases}$ have a nontrivial solution? Why?

Answers: 1. No ($3 \neq 2$); AB is 2×3. 2. $\begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$, trace 0. 3. False (Williams Ex. 2). 4. Yes: homogeneous with 3 unknowns and 2 equations, so infinitely many solutions.

Worked exercises for this lecture (Anton §1.3 and Williams §2.1, original problem numbers) are in the separate file **Lecture3_Solved_Exercises.pdf**. Every number in these notes is checked in **lecture3_matrices.py**.